Solutions / Buns

2110201: COMP ENG MATH (2024/2) Quiz 1

Name: Solutions Student ID: 000000000 Entry Number: 0

1. (1.5 points, 0.5 points each) Answer **True** or **False** for each sub-question in the table provided.

1.1	1.2	1.3
False	True	False

This is in the lecture...

1.1. $G(a, b, c) = (b - c,\ a + 2)$ is a linear transformation.

1.2. Given matrix $\mathbf{T} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $\mathbf{x} = (1, 1)$, and $\mathbf{b} = \mathbf{Tx} = (3, 7)$. We can conclude that (3, 7) is an image of (1, 1) respect to $\mathbf{T}$.

Free 1.3. The vector (1, 1) is **not** an eigenvector of the matrix $\begin{bmatrix} 1 & -1 \\ 2 & -2 \end{bmatrix}$.

$\begin{bmatrix} 1 & -1 \\ 2 & -2 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

eigenvalue $= 0$. (which is okay)

$\therefore \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector.

2. (8.5 points) Given a function f such that

$$f(\underbrace{a, b, c, d}_{\text{4-dimension}}) = \underbrace{(a + 2b - c,\ b + 3c - 4d,\ -a + c - 2d)}_{\text{3 dimension}}$$

$\mathbb{R}^4$ $\mathbb{R}^3$

Let $\mathbf{A}$ be a matrix that represents the linear transformation f.

2.1. (0.5 points) If the transformation f has a domain of $\mathbf{R}^m$ and a range of $\mathbf{R}^n$ (f: $\mathbf{R}^m \Rightarrow \mathbf{R}^n$) where $\mathbf{m}$ and $\mathbf{n}$ are positive integers. Find $\mathbf{m}$ and $\mathbf{n}$.

$m = 4,\ n = 3$

2.2. (2 points) Find the matrix $\mathbf{A}$.

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 3 & -4 \\ -1 & 0 & 1 & -2 \end{bmatrix}$$

2.3. (3 points) Find RRE of the matrix $\mathbf{A}$.

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 3 & -4 \\ -1 & 0 & 1 & -2 \end{bmatrix}$$

$$\xrightarrow{R_3 - R_1} \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 3 & -4 \\ 0 & 2 & 0 & -2 \end{bmatrix}$$

$$\xrightarrow{\frac{1}{2}R_3} \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 3 & -4 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$

$$\xrightarrow[R_2 - R_3]{R_1 - 2R_3} \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 0 & 3 & -3 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$

$$\xrightarrow{\frac{1}{3}R_3} \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$$\xrightarrow{R_1 + R_3} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$$\therefore RRE(A) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix} \quad \#$$

2.4. (2 points) Find all vectors $\mathbf{x}$ such that $\mathbf{Ax} = \mathbf{0}$

$\boxed{\text{1 free column}}$

From $RRE(A) = \begin{array}{c} \begin{matrix} p & p & p & f \end{matrix} \\ \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix} \end{array}$

$h = \begin{pmatrix} p \\ p \\ p \\ f \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$

$\therefore x \in \left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix} s \;\middle|\; s \in \mathbb{R} \right\}$ (set)

or

$x = \begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix} s,\ s \in \mathbb{R}$ (equation)

2.5. (0.5 points) Is the function f one-to-one? (Answer **Yes**/**No** only) No (column 4 has no pivot)

2.6. (0.5 points) Is the function f onto? (Answer **Yes**/**No** only) Yes (every row has a pivot)